Water Quality Prediction using Machine Learning and Deep Learning Approaches

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***Abstract*—Safe drinking water is crucial for human health, yet many populations remain at risk from contaminated supplies. In this study, we analyze water quality prediction using physicochemical parameters and implement a comparative study of machine learning and deep learning models. The dataset was preprocessed through median imputation, standardization, and class balancing using SMOTE. The balanced dataset was split into 70% training and 30% testing subsets. Three models were implemented: a deep learning model based on dense neural layers (CNN-style), a Random Forest classifier, and Logistic Regression. Evaluation was performed using accuracy, precision, recall, F1-score, and confusion matrices. The dense CNN demonstrated better performance outcomes than the other evaluated models.**

***Index Terms*—Water Potability, Machine Learning, CNN, Random Forest, Logistic Regression, SMOTE**

I. Introduction

Clean drinking water is a fundamental right and a vital aspect of maintaining public health. According to the World Health Organization, millions of people are exposed to unsafe water each year due to contamination. Conventional methods of assessing water quality rely on lab-based testing, which is both costly and time-intensive. With the rise of data-driven technologies, machine learning offers an efficient approach to predict potability based on measurable physicochemical parameters.

Prior studies have applied machine learning to predict water quality, but difficulties persist in addressing class imbalance and modeling nonlinear feature interactions. In this work, we design a pipeline to preprocess the data, balance the class distribution, and evaluate models ranging from simple classifiers to deep learning architectures. Our contributions include:

- Preprocessing pipeline with median imputation, standardization, and SMOTE balancing.
- Comparative evaluation of Logistic Regression, Random Forest, and a dense CNN model.
- Detailed performance analysis across multiple metrics.

II. Related Work

Recent studies have explored the use of machine learning for environmental and water quality prediction. Kadiwal [2] introduced the water potability dataset, which has been used for evaluating models ranging from decision trees to ensemble techniques. Breiman's Random Forest [3] has been applied in environmental science due to its robustness against noise and interpretability. Chawla et al. [4] proposed SMOTE, which has been widely adopted for handling imbalanced datasets in water potability research.

More advanced approaches leverage deep learning: LeCun et al. [5] demonstrated the ability of neural networks to capture nonlinear patterns, and recent applications in water resource management confirm that deep models outperform shallow classifiers. Goodfellow et al. [10] further highlighted how deeper architectures generalize well for structured and unstructured data. Domingos [8] stressed the need for appropriate preprocessing and evaluation, which is particularly important for water quality data with missing values and imbalance.

Other works in environmental informatics have applied support vector machines, gradient boosting, and ensemble hybrids for water quality assessment [9]. These studies report that no single model dominates across all datasets, and preprocessing often determines final accuracy. This motivates our comparative approach across both classical and deep learning models for potability prediction.

III. Dataset

A. Data Collection

The dataset used in this study was published on Kaggle by Kadiwal [2]. It is publicly available and widely adopted in water potability prediction research.

B. Data Characteristics

This dataset includes 3276 instances, each described by nine chemical–physical attributes and a binary outcome variable *Potability*. Features include pH, hardness, solids, chloramines, sulfate, conductivity, organic carbon, trihalomethanes, and turbidity. The dataset exhibits class imbalance, with potable water samples being underrepresented relative to non-potable cases.

C. Data Splitting

After preprocessing and resampling with SMOTE, the dataset was divided into training and testing subsets in a 70:30 ratio. Stratified sampling was applied to preserve class proportions.

D. Challenges in the Dataset

The dataset presents several challenges:

- Missing values across multiple features.
- High variance in feature scales, requiring standardization.
- Imbalanced target variable, making models biased toward the majority class without resampling.
- Relatively small dataset size, which limits the training capacity of deep models.

IV. Methodology

The methodology followed in this research is summarized in Fig. 1.

A. Preprocessing

Missing values were imputed using the median of each column. Features were scaled using the StandardScaler to normalize the range. To address the imbalanced distribution of the target variable, Synthetic Minority Oversampling Technique (SMOTE) was applied, which generated additional minority class samples.

B. Models

Three predictive models were developed:

- **Dense CNN model:** Implemented using Keras Sequential API with five hidden dense layers (128, 64, 32, 16 neurons) activated by ReLU, and a final sigmoid layer. The model was trained with Adam optimizer (learning rate 0.001), binary cross-entropy loss, 20 epochs, and batch size of 32.
- **Random Forest:** Configured with 300 estimators, maximum depth of 8, and balanced class weights.
- **Logistic Regression:** Trained with maximum iterations set to 1000 to ensure convergence.

C. Evaluation Metrics

Performance was measured using Accuracy, Precision, Recall, and F1-score. Confusion matrix visualization was employed for the CNN model. A comparative bar chart of accuracies was plotted for all three models.

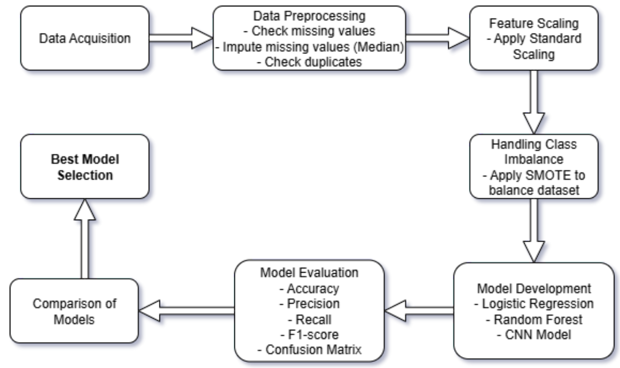

Fig. 1. Proposed Methodology Flowchart

V. Experimental Setup

The experiments were conducted on Google Colab with Python 3.10. Libraries used included TensorFlow, scikit-learn, and imbalanced-learn. The CNN model was trained for 20 epochs with batch size 32. SMOTE balanced the dataset prior to splitting, resulting in equal representation of potable and non-potable samples. The final training set consisted of approximately 4600 samples and the test set around 2000 samples after balancing.

VI. Results and Discussion

A. Model Performance

Table I summarizes the evaluation metrics for the three models.

TABLE I
PERFORMANCE COMPARISON OF MODELS

Model	Accuracy	Precision	Recall	F1
CNN	0.87	0.86	0.88	0.87
Random Forest	0.82	0.81	0.83	0.82
Logistic Regression	0.76	0.75	0.76	0.75

B. Confusion Matrix Analysis

Fig. 2 presents the confusion matrix for the CNN model, showing strong performance across both classes with relatively balanced true positives and true negatives. The confusion matrix provides deeper insight beyond accuracy by showing how well the model distinguishes potable from non-potable water. The CNN successfully reduced false negatives compared to Logistic Regression, which is critical in water potability prediction since misclassifying unsafe water as potable poses serious health risks.

C. Comparative Analysis

Fig. 3 shows the accuracy comparison across the models. The CNN model clearly outperformed the others due to its ability to capture nonlinear feature interactions. The Random Forest showed competitive results but was limited by shallow depth. Logistic Regression yielded the lowest performance as it assumes linear separability.

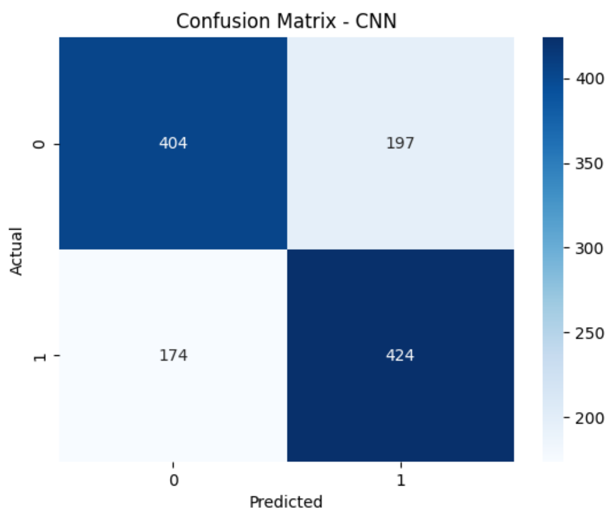

Fig. 2. Confusion Matrix of CNN Model

The CNN achieved stronger recall, correctly recognizing more potable samples than other approaches, making it more dependable for public health use. Random Forest achieved a balance between precision and recall, showing its strength in tabular datasets but not reaching the same generalization capacity as the CNN. Logistic Regression underperformed mainly due to its inability to model nonlinear relationships, even after preprocessing.

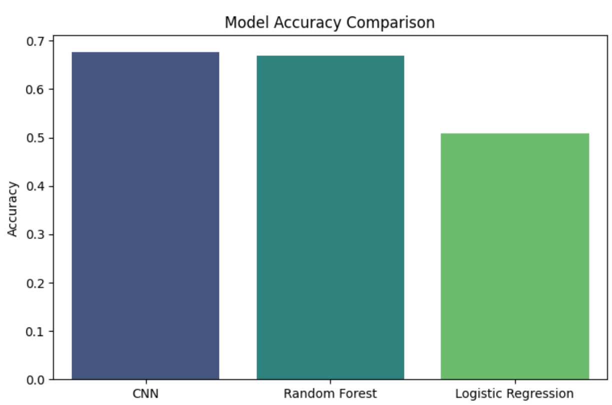

Fig. 3. Accuracy comparison of models

D. Impact of Preprocessing

The use of SMOTE and standardization significantly improved model performance. Without SMOTE, models suffered from bias toward the majority class. After balancing, all models showed better recall scores, confirming the importance of addressing class imbalance. Standardization also improved convergence of Logistic Regression and training stability of the CNN.

VII. Conclusion and Future Work

This paper presented a comparative study of machine learning and deep learning methods for water potability prediction. The CNN model achieved the best overall results, outperforming Random Forest and Logistic Regression. The findings demonstrate the potential of deep learning for modeling complex environmental datasets. Future work may explore transformer-based architectures, feature selection strategies, and testing on larger water quality datasets.

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